05. Train-Test Split & Validation Strategy

Understand the split and what a robust evaluation should look like.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

Notebook split

The notebook uses train_test_split(X, y, test_size=0.2, random_state=42), resulting in an 80:20 split. It does not set stratify=y, so the class proportions can differ between train and test. It repeats the same split after feature engineering, again with the same seed.

Leakage concerns

- All missing-value imputation is fitted on the full dataset before splitting. Test-set information influences imputation statistics.
- Missing target values are imputed before splitting, contaminating both train and test labels with fabricated outcomes.
- OneHotEncoder is fitted on the full data before splitting, so category vocabulary is informed by the test set. This is a mild form of preprocessing leakage.
- The scaler is correctly fitted only on X_train in each modelling pass.

Recommended evaluation design

- First remove rows whose target is unknown; preserve them for separate scoring only if labels become available later.
- Create a stratified train/test split using stratify=y, preferably with a fixed random seed.
- Build all imputers, encoders, scalers and feature transformations inside a Pipeline/ColumnTransformer.
- Use stratified k-fold cross-validation on training data for model selection and tuning.
- Keep the test set untouched until model/threshold decisions are complete; evaluate it once.
- If data includes application dates, prefer a temporal holdout to test future performance.

Metrics and decision threshold

Report confusion matrix, precision, recall, F1, ROC-AUC and PR-AUC. Evaluate calibration (Brier score/reliability curve) if predicted probabilities inform risk decisions. Choose threshold based on the costs of false approval and false rejection, policy constraints and fairness review—not a default 0.5 alone.

Interview response

> “The notebook uses an 80/20 random split with a fixed seed, but it does not stratify. More importantly, preprocessing is performed before splitting, so the imputation and encoding have seen the test data. I would move preprocessing into a pipeline, exclude missing target labels and use stratified cross-validation or a temporal holdout.”